CSCE 314 Programming Languages

Haskell: Types, Currying, and Polymorphism

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Types

A <u>type</u> is a collection of related values. For example,

- Bool contains the two logical values True and False
- Int contains values $-2^{63}, \ldots, -1, 0, 1, \ldots, 2^{63} - 1$

If evaluating an expression e would produce a value of type T, then e <u>has type</u> T, written

$$\boxed{e :: T}$$

Every well-formed expression has a type, which can be automatically calculated at *compile time* using a process called <u>type inference</u>

Type Errors

Applying a function to one or more arguments of the wrong type is called a <u>type error</u>

```
> 1 + False
Error
```

1 is a number and False is a logical value, but + requires two numbers

<u>Static type</u> checking - all type errors are found at <u>compile time</u>, which makes programs safer and faster by removing the need for type checks at run time

Type Annotations

Programmer can (and at times must) annotate expressions with type in the form `e :: T`

For example,

- True :: Bool
- 5 :: Int -- type is really (Num t) => t
- (5 + 5) :: Int -- likewise
- (7 < 8) :: Bool

Some expressions can have many types, e.g.,

5 :: Int, 5 :: Integer, 5 :: Float

GHCi command :type e shows the type of (the result of) e

```
> not False
True

> :type not False
not False :: Bool
```

Basic Types

Haskell has a number of basic types, including:

- `Bool` - logical values
- `Char` - single characters
- `String` - lists of characters type String = [Char]
- `Int` - fixed-precision integers
- `Integer` - arbitrary-precision integers
- `Float` - single-precision floating-point numbers
- `Double` - double-precision floating-point numbers

List Types

A <u>list</u> is sequence of values of the <u>same</u> type:

```
[False,True,False] :: [Bool]
['a','b','c']  :: [Char]
"abc" :: [Char]
[[True, True], []] :: [[Bool]]
```

Note:

- [t] has the type list with elements of type t
- The type of a list says nothing about its length
- The type of the elements is unrestricted
- Lists can be infinite: l = [1..]

Tuple Types

A <u>tuple</u> is a sequence of values of <u>different</u> types:

```
(False,True)       :: (Bool,Bool)
(False,'a',True)   :: (Bool,Char,Bool)
("Howdy",(True,2)) :: ([Char],(Bool,Int))
```

Note:

- (t1,t2,…,tn) is the type of n-tuples whose i-th component has type ti for any i in 1…n
- The type of a tuple encodes its size
- The type of the components is unrestricted
- Tuples with arity one are not supported: (t) is parsed as t, parentheses are ignored

Function Types

A <u>function</u> is a mapping from values of one type (T1) to values of another type (T2), with the type T1 -> T2

```
not     :: Bool -> Bool
isDigit :: Char -> Bool
toUpper :: Char -> Char
(&&) :: Bool -> Bool -> Bool
```

Note:
The argument and result types are unrestricted. Functions with multiple arguments or results are possible using lists or tuples:

```
add     :: (Int,Int) → Int
add (x,y) = x+y

zeroto :: Int → [Int]
zeroto n  = [0..n]
```

One parameter functions!

Curried Functions

Functions with multiple arguments are also possible by returning functions as results:

```
add :: (Int,Int) → Int
add (x,y)  = x+y

add' :: Int → (Int → Int)
add' x y = x+y
```

add' takes an int x and returns a function add' x. In turn, this function takes an int y and returns the result x+y

Note:

- add and add' produce the same final result, but add takes its two arguments at the same time, whereas add' takes them one at a time
- Functions that take their arguments one at a time are called curried functions, celebrating the work of Haskell Curry on such functions

Curried Functions (Cont.)

Functions with more than two arguments can be curried by returning nested functions:

```
mult      :: Int → (Int → (Int → Int))
mult x y z = x*y*z
```

mult takes an integer x and returns a function <u>mult x</u>, which in turn takes an integer y and returns a function <u>mult x y</u>, which finally takes an integer z and returns the result x*y*z

Note:

- Functions returning functions: an example of higher-order functions
- Unless tupling is explicitly required, all functions in Haskell are normally defined in curried form

Why is Currying Useful?

Curried functions are more flexible than functions on tuples, because useful functions can often be made by <u>partially applying</u> a curried function

For example:

```
add' 1 :: Int -> Int
take 5 :: [a] -> [a]
drop 5 :: [a] -> [a]
```

```
map :: (a->b) -> [a] -> [b]
map f  []     = []
map f (x:xs) = f x : map f xs

> map (add' 1) [1,2,3]
[2,3,4]
```

Currying Conventions

To avoid excess parentheses when using curried functions, two simple conventions are adopted:

1. The arrow → (type constructor) associates to the <u>right</u>

```
Int → Int → Int → Int
```

Means Int → (Int → (Int → Int))

2. As a consequence, it is then natural for function application to associate to the <u>left</u>

```
mult x y z
```

Means ((mult x) y) z

Polymorphic Functions

A function is called <u>polymorphic</u> ("of many forms") if its type contains one or more type variables Thus, polymorphic functions work with many types of arguments

```
length :: [a] → Int
```

for any type a, length takes a list of values of type a and returns an integer

```
id :: a → a
```

for any type a, id maps a value of type a to itself

```
head :: [a] → a
take :: Int→[a]→[a]
```

a is a type variable

Polymorphic Types

Type variables can be instantiated to different types in different circumstances:

```
> length [False,True]
2
> length [1,2,3,4::Int]
4
```

a = Bool

a = Int

expression	polymorphic type	type variable bindings	resulting type
`id`	`a -> a`	`a=Int`	`Int -> Int`
`id`	`a -> a`	`a=Bool`	`Bool -> Bool`
`length`	`[a] -> Int`	`a=Char`	`[Char] -> Int`
`fst`	`(a, b) -> a`	`a=Char, b=Bool`	`Char`
`snd`	`(a, b) -> b`	`a=Char, b=Bool`	`Bool`
`([], [])`	`([a], [b])`	`a=Char, b=Bool`	`([Char], [Bool])`

Type variables must begin with a lower-case letter, and are usually named a, b, c, etc.

Overloaded Functions

A polymorphic function is called <u>overloaded</u> if its type contains one or more class constraints

```
sum :: Num a ⇒ [a] → a
```

for any numeric type a, sum takes a list of values of type a and returns a value of type a

Constrained type variables can be instantiated to any types that satisfy the constraints:

```
> sum [1,2,3]
6
> sum [1.1,2.2,3.3]
6.6
> sum ['a','b','c']
ERROR
```

a = Int

a = Float

Char is not a numeric type

Class Constraints

Recall that polymorphic types can be instantiated with all types, e.g.,

```
id :: t -> t    length :: [t] -> Int
```

This is when no operation is subjected to values of type `t`

What are the types of these functions?

```
min :: Ord a => a -> a -> a
min x y = if x < y then x else y

elem :: Eq a => a -> [a] -> Bool
elem x (y:ys) | x == y = True
elem x (y:ys) = elem x ys
elem x [] = False
```

Type variables can only be bound to types that satisfy the constraints

`Ord a` and `Eq a` are class constraints

Type Classes

Constraints arise because values of the generic types are subjected to operations that are not defined for all types:

```
min :: Ord a => a -> a -> a
min x y = if x < y then x else y

elem :: Eq a => a -> [a] -> Bool
elem x (y:ys) | x == y = True
elem x (y:ys) = elem x ys
elem x [] = False
```

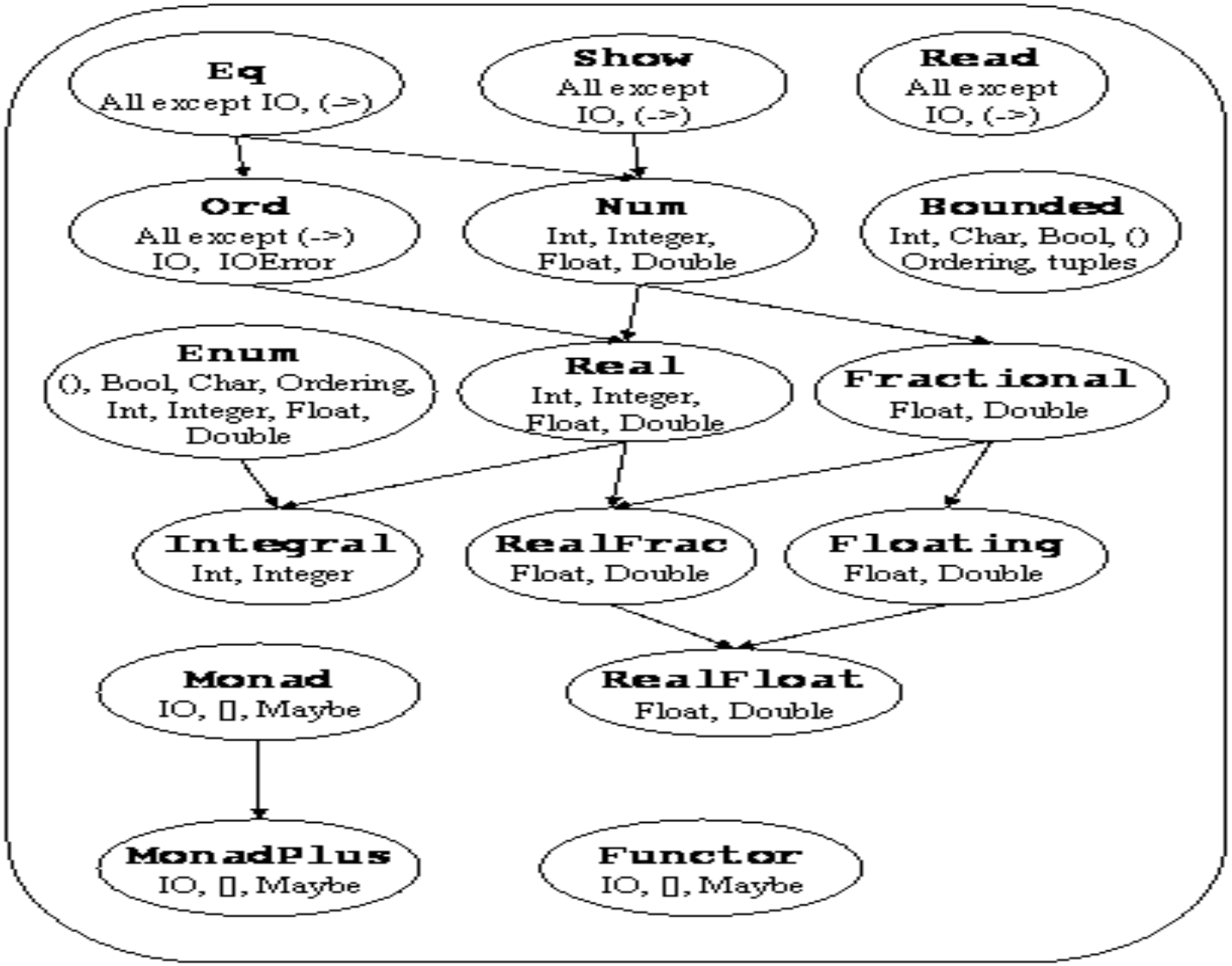
`Ord` and `Eq` are type classes:

- Num (Numeric types)
- Eq (Equality types)
- Ord (Ordered types)

```
(+)  :: Num a ⇒ a → a → a
(==) :: Eq a  ⇒ a → a → Bool
(<)  :: Ord a ⇒ a → a → Bool
```

Haskell 98 Class Hierarchy

For detailed explanation, refer
http://www.haskell.org/onlinereport/basic.html



The Eq and Ord Classes

```
class  Eq a  where
  (==), (/=)  ::  a -> a -> Bool
  x /= y  = not (x == y)
  x == y  = not (x /= y)
class  (Eq a) => Ord a  where
  compare  :: a -> a -> Ordering
  (<), (<=), (>=), (>) :: a -> a -> Bool
  max, min  :: a -> a -> a

  compare x y | x == y  = EQ
              | x < y   = LT
              | otherwise = GT
```

```
x <= y = compare x y /= GT
x <  y = compare x y == LT
x >= y = compare x y /= LT
x >  y = compare x y == GT
```

```
max x y | x <= y   =  y
        | otherwise =  x
min x y | x <= y   =  x
        | otherwise =  y
```

The Enum Class

```
class  Enum a  where
  toEnum     :: Int -> a
  fromEnum  :: a -> Int
  succ, pred  :: a -> a
   . . .
-- Minimal complete definition: toEnum, fromEnum
```

Note: these methods only make sense for types that map injectively into Int using fromEnum and toEnum

```
succ  =  toEnum . (+1) . fromEnum
pred  =  toEnum . (subtract 1) . fromEnum
```

The Show and Read Classes

```
class Show a where
  show :: a -> String
```

```
class Read a where
  read :: String -> a
```

Many types are showable and/or readable

```
> show 10
"10"
> show [1,2,3]
"[1,2,3]"
```

```
> read "10" :: Int
10
> read "[1,2,3]" :: [Int]
[1,2,3]
```

```
> map (* 2.0) (read "[1,2]")
[2.0,4.0]
```

Hints and Tips

When defining a new function in Haskell, it is useful to begin by writing down its type

Within a script, it is good practice to state the type of every new function defined

When stating the types of polymorphic functions that use numbers, equality or orderings, take care to include the necessary class constraints

Exercises

(1) What are the types of the following values?

```
['a','b','c']

('a','b','c')

[(False,'0'),(True,'1')]

([False,True],['0','1'])

[tail,init,reverse]
```

(2) What are the types of the following functions?

```
second xs          = head (tail xs)

swap (x,y)         = (y,x)

pair x y           = (x,y)

double x           = x*2

palindrome xs      = reverse xs == xs

twice f x = f (f x)

lessThanHalf x y = x * 2 < y
```

(3) Check your answers using GHCi.